



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



GNSS Spoofing, Jamming, and Satellite Dependency Risk for Aviation Operators

Threat Report

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TLP:CLEAR | Lost Boys Cyber | GNSS Spoofing, Jamming, and Satellite Dependency Risk for Aviation Operators - Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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GNSS Spoofing, Jamming, and Satellite Dependency Risk for Aviation Operators

1. Executive Summary

Aviation cyber reporting continues to highlight GNSS spoofing/jamming and satellite-dependent operational dependencies as cyber-hybrid risks requiring monitoring and contingency procedures. Lost Boys Cyber selected this item for daily analyst review because it affects high-value defender decisions around exposure reduction, identity control, endpoint monitoring, and incident readiness.

This report is written for security leaders and SOC operators who need an actionable, source-backed view rather than a news summary. It includes source confidence, likely TTPs, detection logic, KQL hunts, risk assessment, and concrete defender actions.

2. Intelligence Requirements

Question	Current Answer	Confidence
What is the core activity?	Aviation cyber reporting continues to highlight GNSS spoofing/jamming and satellite-dependent operational dependencies as cyber-hybrid risks requiring monitoring and contingency procedures.	Medium-High
Who should prioritize it?	Airlines, airports, ANSPs, satellite communications, flight operations	Medium
What should defenders do today?	Validate exposure, run hunts, review privileged access, and prepare containment actions.	High

3. Source Review & Confidence

Source	Contribution	Confidence
GBHackers aviation/aerospace attacks overview	Highlights GNSS spoofing/jamming, satellite communications, SaaS, and shared-platform risks.	Medium
EASA cybersecurity topic page	Regulatory safety/security context for aviation cybersecurity risk management.	High
CISA GPS/GNSS resilience resources	Government resilience context for positioning, navigation, and timing dependencies.	High

Source depth was sufficient for an analyst-facing threat report. Where reporting is based on vendor or media summaries rather than primary telemetry, claims are framed as reported and should be validated against local evidence.

4. Threat Activity Overview

The activity centers on GNSS spoofing, jamming, SATCOM, PNT dependencies. The operational pattern is consistent with modern intrusion operations: obtain access through exposed services, stolen credentials, phishing, or third-party platforms; convert access into persistence or data access; then monetize through espionage, fraud, credential resale, or ransomware/extortion.

5. Affected Sectors and Exposure

Exposure Area	Why It Matters
Airlines, airports, ANSPs, satellite communications, flight operations	Primary audience for prioritization and control validation.
Identity and remote access	Most listed scenarios become serious when credentials, VPNs, RMM, or admin portals are abused.
Endpoint and server telemetry	Required to distinguish isolated alerts from intrusion chains.
Third-party platforms	Shared platforms can create cascading exposure even when the defended network is segmented.

6. Aviation / Aerospace Sector Impact

Field	Assessment
Affected Segment	Airlines, airports, ANSPs, satellite communications, flight operations
Operational Relevance	Flight operations, dispatch, navigation awareness, weather/track feeds, and airport surface operations should validate contingency playbooks for loss or manipulation of trusted timing/location feeds.
Likely Exposure Paths	Identity compromise, vendor remote access, unmanaged SaaS integrations, exposed portals, and third-party support workflows.
Third-Party / Supply-Chain Considerations	Require supplier notification SLAs, evidence of MFA/EDR coverage, backup testing, and segmentation between enterprise IT and operational aviation workflows.
Defender Actions	Review access logs, validate backup/restore plans, tighten conditional access, and coordinate with sector ISAC / government guidance where applicable.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed or Plausible Technique	Evidence
Initial Access	T1190 Exploit Public-Facing Application / T1566 Phishing / T1078 Valid Accounts	Selected based on reporting around exposed services, identity intrusion, or malware delivery.
Execution	T1059 Command and Scripting Interpreter	Common follow-on for loaders, RATs, ransomware staging, and administrative abuse.
Credential Access	T1003 OS Credential Dumping / credential harvesting	Relevant where credential-theft or ransomware access brokerage is reported.
Command and Control	T1102 Web Service / T1071 Application Layer Protocol	Relevant where Telegram, cloud services, web panels, or SaaS channels are noted.
Impact	T1486 Data Encrypted for Impact / T1489 Service Stop	Relevant for ransomware and extortion-linked activity.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
GNSS spoofing, jamming, SATCOM, PNT dependencies	Tooling / malware / infrastructure theme	Use as keyword pivots in EDR, proxy, SIEM, ticket, and threat-intel systems.
Newly created privileged sessions	Behavioral indicator	Detect compromised admin, VPN, RMM, or cloud accounts.
Archive/compression/encryption prior to outbound transfer	Behavioral indicator	Detect data staging before exfiltration or extortion.
Unusual SaaS/cloud channel use by servers	Behavioral indicator	Detect C2 or covert transfer using legitimate services.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Keyword/tool name appears in command line, alert title, file path, or threat-intel note	EDR/SIEM	Fast triage pivot for this report's named activity.
Rare outbound connection from server to consumer/cloud messaging service	Network/proxy/DNS	Identify C2 or exfiltration using legitimate services.
Privileged account used from new ASN/device followed by process creation	IAM + EDR	Detect identity-to-endpoint intrusion chaining.

9.2. Sigma / Detection Content

<pre> title: Lost Boys Cyber - GNSS Spoofing, Jamming, and Satellite Dependency Risk for Aviation Ope id: lbc-gnss-spoofing-jamming-and-satellite-dependency-ris status: experimental description: Behavioral keyword and post-exploitation detection for the daily threat item GNSS Spoofing, Jamming, and Satellite Dependency Risk for Aviation Operators. logsource: product: windows category: process_creation detection: selection_keywords: CommandLine contains: - 'GNSS spoofing' - 'ransomware' - 'stealer' - 'credential' selection_tools: Image endswith: - '\powershell.exe' - '\cmd.exe' - '\rundll32.exe' - '\wscript.exe' - '\7z.exe' - '\rar.exe' condition: selection_keywords or selection_tools falsepositives: - Security testing, malware analysis, or administrator troubleshooting level: medium </pre>
--

9.3. Hunt Query

```
DeviceProcessEvents
| where Timestamp > ago(30d)
| where ProcessCommandLine has_any ('GNSS spoofing', 'jamming', 'SATCOM', 'PNT dependencies')
  or FileName in~
('powershell.exe', 'cmd.exe', 'rundll32.exe', 'wscript.exe', 'cscript.exe', '7z.exe', 'rar.exe')
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Hosts=dcount(DeviceName),
Examples=make_set(ProcessCommandLine, 5) by FileName, InitiatingProcessFileName
| order by LastSeen desc
```

10. Threat Hunting

10.1. Hunt Hypothesis

If this activity affected the environment, defenders should see a chain that connects exposure or identity abuse to suspicious endpoint execution, outbound network activity, archive creation, or ransomware/extortion preparation.

10.2. Hunt Query

```
let keywords = dynamic(['GNSS spoofing', 'jamming', 'SATCOM', 'PNT dependencies']);
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where RemoteUrl has_any (keywords) or InitiatingProcessCommandLine has_any (keywords)
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, RemoteUrl,
RemoteIP, RemotePort
| union (
    DeviceFileEvents
    | where Timestamp > ago(30d)
    | where FileName endswith '.7z' or FileName endswith '.rar' or FileName endswith '.zip' or
FolderPath has_any (keywords)
    | project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine,
RemoteUrl='', RemoteIP='', RemotePort=int(null)
)
```

11. Risk Assessment

Dimension	Assessment
Likelihood	Medium where relevant exposure exists; lower where controls and segmentation are mature.
Impact	Credential theft, data exposure, operational disruption, ransomware, or third-party impact depending on scenario.
Confidence	Medium overall; higher for named source claims and lower for local environment applicability.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC	Run the provided hunts and review any matches with identity/network context.
High	IT / IAM	Enforce MFA, rotate suspect credentials, and restrict admin interfaces.
High	Vulnerability / Platform Owners	Patch or harden exposed systems tied to this activity.
Medium	IR	Confirm backups, escalation contacts, and

		evidence-preservation steps.
Medium	Threat Intel	Track source updates and update detections if atomic IOCs become available.

13. References

- [1] GBHackers aviation/aerospace attacks overview. <https://gbhackers.com/attacks-on-aviation-and-aerospace/>
- [2] EASA cybersecurity topic page. <https://www.easa.europa.eu/en/domains/cybersecurity>
- [3] CISA GPS/GNSS resilience resources. <https://www.cisa.gov/resources-tools/resources/resilient-positioning-navigation-and-timing-pnt-reference-architecture>